Linear Regression

Problem Setup

Let y be an explained variable and x an explanatory variable related by $y = f(x)$. The linear model assumes $y = a + bx$, where $a, b \in \mathbb{R}$. This relationship does not hold exactly, but may hold up to some noise ε, a term capturing random processes we do not fully understand. We write:

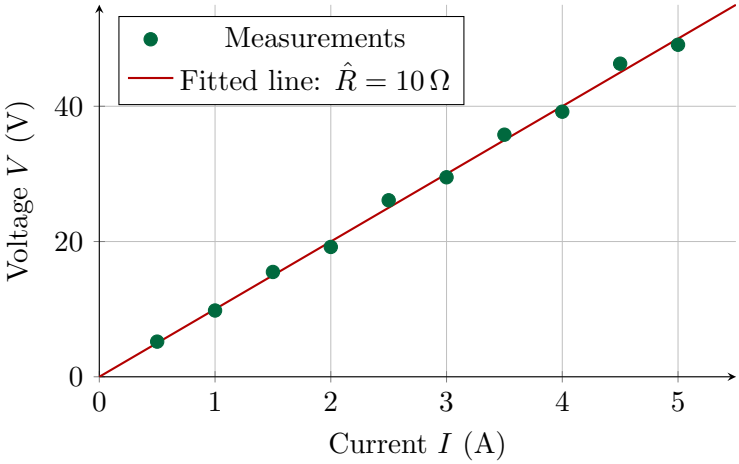
$$Y = a + bX + \varepsilon$$

The goal is to find estimators $\hat{a}, \hat{b}$ of the true parameters a, b. The case $x \in \mathbb{R}$ is called univariate regression; the case $x \in \mathbb{R}^n$ with $n \geq 2$ is called multivariate regression.

Practical Example: Estimating Resistance via Ohm's Law

Ohm's Law states that voltage V and current I are related by $V = IR$, where R is the resistance. Given measurements $(I_1, V_1), \ldots, (I_n, V_n)$ from a circuit, we can estimate R by fitting a line through the origin.

Ohm's Law: Estimating Resistance



The slope of the fitted line gives the resistance R. This is a direct application of linear regression: the current I is the explanatory variable, the voltage V is the explained variable, and the unknown parameter R is what we seek to estimate.

Choosing the Loss

To find estimators $\hat{a}, \hat{b}$, we need a notion of "goodness of fit" - a loss function measuring how far our predictions are from the observed values.

Geometric Interpretation

Given a point (x_i, y_i) and a candidate line $\hat{y} = a + bx$, there are several ways to measure the "distance" from the point to the line:

- **Vertical distance**: $|y_i - (a + bx_i)|$ - the difference in y-coordinates
- **Horizontal distance**: difference in x-coordinates
- **Euclidean distance**: perpendicular distance to the line

In regression, we use the **vertical distance**. This is natural: we are trying to predict y from x, so the error is how far off our prediction $\hat{y}_i = a + bx_i$ is from the observed y_i.

Squaring these vertical distances gives the squared loss:

$$\ell(y, \hat{y}) = (y - \hat{y})^2$$

Theoretical Justification: Bayes Optimality

Squared loss has a deeper justification. Under squared loss $\ell(y, z) = (y - z)^2$, the Bayes optimal predictor - the function minimizing expected risk - is the conditional expectation $\mathbb{E}[y|x]$. Intuitively, $\mathbb{E}[y|x = x']$ answers: *what do we expect y to be, given we observed $x = x'$?*

To show this, let y be the target, x the input, and $z \in \mathbb{R}$ a candidate prediction. The optimal predictor satisfies:

$$f_*(x') \in \arg\min_{z \in \mathbb{R}} \mathbb{E}\left[(y - z)^2 | x = x'\right] \tag{1}$$

$$= \arg\min_{z \in \mathbb{R}} \left\{\mathbb{E}\left[(y - \mathbb{E}[y|x = x'])^2 | x = x'\right] + (z - \mathbb{E}[y|x = x'])^2\right\} \tag{2}$$

The first term is the conditional variance, independent of z. The second term is minimized at $z = \mathbb{E}[y|x = x']$. Therefore:

$$f_*(x') = \mathbb{E}[y|x = x']$$

The Bayes risk $\mathcal{R}^*$ is the expected conditional variance. To see this, substitute f_* into the risk:

$$\mathcal{R}^* = \mathbb{E}[(y - f_*(x))^2] = \mathbb{E}[(y - \mathbb{E}[y|x])^2] = \mathbb{E}_x\left[\mathrm{Var}(y|x)\right]$$

This is the irreducible error: even with the optimal predictor, we cannot do better than the inherent noise in y given x.

When we fit a linear model $\hat{y} = a + bx$, we approximate $\mathbb{E}[y|x]$ with the best linear function. If the true conditional expectation is linear in x, the model is correctly specified; otherwise, we incur approximation error.

Deriving the OLS Estimators

Consider the univariate case. Let $(x_i, y_i) \in \mathcal{X} \times \mathcal{Y}$ be observations sampled from an unknown distribution ν, with training set $D_n = \{(x_i, y_i)\}_{i=1}^n$. The goal is to predict $y \in \mathcal{Y}$ given a new input $x \in \mathcal{X}$.

Under squared loss $\ell = (y - \hat{f})^2$, we minimize the empirical risk:

$$\hat{\mathcal{R}}_n(f) = \frac{1}{n}\sum_{i=1}^n (y_i - f(x_i))^2$$

For the linear model $f(x) = a + bx$, we seek $a, b \in \mathbb{R}$ minimizing $\hat{\mathcal{R}}_n$. We differentiate with respect to each parameter and set to zero.

Derivation of $\hat{a}$ (Intercept)

We compute $\frac{\partial \hat{\mathcal{R}}_n}{\partial a}$:

$$\begin{aligned}
\frac{\partial \hat{\mathcal{R}}_n}{\partial a} &= \frac{\partial}{\partial a}\left[\frac{1}{n}\sum_{i=1}^n (y_i - f(x_i))^2\right] && (3)\\
&= \frac{\partial}{\partial a}\left[\frac{1}{n}\sum_{i=1}^n (y_i - a - bx_i)^2\right] && (4)\\
&= \frac{1}{n}\sum_{i=1}^n \frac{\partial}{\partial a}(y_i - a - bx_i)^2 && (5)\\
&= \frac{1}{n}\sum_{i=1}^n 2(y_i - a - bx_i)\cdot\frac{\partial}{\partial a}(y_i - a - bx_i) && (6)\\
&= \frac{1}{n}\sum_{i=1}^n 2(y_i - a - bx_i)\cdot(-1) && (7)\\
&= -\frac{2}{n}\sum_{i=1}^n (y_i - a - bx_i) && (8)\\
&= \frac{2}{n}\sum_{i=1}^n (a + bx_i - y_i) && (9)
\end{aligned}$$

Setting this to zero:

$$\begin{aligned}
0 &= \frac{2}{n}\sum_{i=1}^n (a + bx_i - y_i) && (10)\\
0 &= \sum_{i=1}^n (a + bx_i - y_i) && (11)\\
0 &= \sum_{i=1}^n a + \sum_{i=1}^n bx_i - \sum_{i=1}^n y_i && (12)
\end{aligned}$$

$$0 = na + b\sum_{i=1}^{n} x_i - \sum_{i=1}^{n} y_i \tag{13}$$

$$na = \sum_{i=1}^{n} y_i - b\sum_{i=1}^{n} x_i \tag{14}$$

$$\hat{a} = \frac{1}{n}\sum_{i=1}^{n} y_i - b \cdot \frac{1}{n}\sum_{i=1}^{n} x_i \tag{15}$$

$$\hat{a} = \bar{y} - b\bar{x} \tag{16}$$

Here $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$ and $\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$ are the sample means.

$$\boxed{\hat{a} = \bar{y} - \hat{b}\bar{x}}$$

Derivation of $\hat{b}$ (Slope)

We compute $\frac{\partial \hat{\mathcal{R}}_n}{\partial b}$:

$$\frac{\partial \hat{\mathcal{R}}_n}{\partial b} = \frac{\partial}{\partial b}\left[\frac{1}{n}\sum_{i=1}^{n}(y_i - a - bx_i)^2\right] \tag{17}$$

$$= \frac{1}{n}\sum_{i=1}^{n}\frac{\partial}{\partial b}(y_i - a - bx_i)^2 \tag{18}$$

$$= \frac{1}{n}\sum_{i=1}^{n} 2(y_i - a - bx_i) \cdot \frac{\partial}{\partial b}(y_i - a - bx_i) \tag{19}$$

$$= \frac{1}{n}\sum_{i=1}^{n} 2(y_i - a - bx_i) \cdot (-x_i) \tag{20}$$

$$= -\frac{2}{n}\sum_{i=1}^{n} x_i(y_i - a - bx_i) \tag{21}$$

$$= \frac{2}{n}\sum_{i=1}^{n}(bx_i^2 - x_iy_i + ax_i) \tag{22}$$

Setting this to zero:

$$\frac{2}{n}\sum_{i=1}^{n}(bx_i^2 - x_iy_i + ax_i) = 0 \tag{23}$$

$$\sum_{i=1}^{n}(bx_i^2 - x_iy_i + ax_i) = 0 \tag{24}$$

$$b\sum_{i=1}^{n} x_i^2 - \sum_{i=1}^{n} x_iy_i + a\sum_{i=1}^{n} x_i = 0 \tag{25}$$

Substituting $\hat{a} = \bar{y} - b\bar{x}$:

$$b\sum_{i=1}^{n} x_i^2 - \sum_{i=1}^{n} x_i y_i + (\bar{y} - b\bar{x})\sum_{i=1}^{n} x_i = 0 \tag{26}$$

$$b\sum_{i=1}^{n} x_i^2 - \sum_{i=1}^{n} x_i y_i + \bar{y}\sum_{i=1}^{n} x_i - b\bar{x}\sum_{i=1}^{n} x_i = 0 \tag{27}$$

$$b\sum_{i=1}^{n} x_i^2 - b\bar{x}\sum_{i=1}^{n} x_i = \sum_{i=1}^{n} x_i y_i - \bar{y}\sum_{i=1}^{n} x_i \tag{28}$$

$$b\left[\sum_{i=1}^{n} x_i^2 - \bar{x}\sum_{i=1}^{n} x_i\right] = \sum_{i=1}^{n} x_i y_i - \bar{y}\sum_{i=1}^{n} x_i \tag{29}$$

$$b\left[\sum_{i=1}^{n} x_i^2 - \frac{1}{n}\sum_{i=1}^{n} x_i \sum_{i=1}^{n} x_i\right] = \sum_{i=1}^{n} x_i y_i - \frac{1}{n}\sum_{i=1}^{n} y_i \sum_{i=1}^{n} x_i \tag{30}$$

$$\hat{b} = \frac{\sum_{i=1}^{n} x_i y_i - \frac{1}{n}\sum_{i=1}^{n} y_i \sum_{i=1}^{n} x_i}{\sum_{i=1}^{n} x_i^2 - \frac{1}{n}\sum_{i=1}^{n} x_i \sum_{i=1}^{n} x_i} \tag{31}$$

Simplifying the numerator (sample covariance):

$$\sum_{i=1}^{n} x_i y_i - \frac{1}{n}\sum_{i=1}^{n} y_i \sum_{i=1}^{n} x_i \tag{32}$$

$$= \sum_{i=1}^{n} x_i y_i - \bar{y}\sum_{i=1}^{n} x_i - \bar{x}\sum_{i=1}^{n} y_i + n\frac{1}{n}\sum_{i=1}^{n} y_i \frac{1}{n}\sum_{i=1}^{n} x_i \tag{33}$$

$$= \sum_{i=1}^{n} x_i y_i - \bar{y}\sum_{i=1}^{n} x_i - \bar{x}\sum_{i=1}^{n} y_i + n\bar{y}\bar{x} \tag{34}$$

$$= \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y}) \tag{35}$$

Simplifying the denominator (sample variance):

$$\sum_{i=1}^{n} x_i^2 - \frac{1}{n}\sum_{i=1}^{n} x_i \sum_{i=1}^{n} x_i \tag{36}$$

$$= \sum_{i=1}^{n} x_i^2 - 2\frac{1}{n}\sum_{i=1}^{n} x_i \sum_{i=1}^{n} x_i + n\frac{1}{n}\sum_{i=1}^{n} x_i \frac{1}{n}\sum_{i=1}^{n} x_i \tag{37}$$

$$= \sum_{i=1}^{n} x_i^2 - 2\bar{x}\sum_{i=1}^{n} x_i + n\bar{x}^2 \tag{38}$$

$$= \sum_{i=1}^{n} (x_i - \bar{x})(x_i - \bar{x}) \tag{39}$$

$$\boxed{\hat{b} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2} = \frac{\widehat{\mathrm{Cov}}(X,Y)}{\widehat{\mathrm{Var}}(X)}}$$

Properties of the Residuals

The first-order conditions impose constraints on the residuals $\varepsilon_i = y_i - \hat{a} - \hat{b}x_i$.

Zero Mean

From the first-order condition for a:

$$\frac{2}{n}\sum_{i=1}^{n}(a + bx_i - y_i) = 0$$

Since $\varepsilon_i = y_i - a - bx_i$:

$$\frac{2}{n}\sum_{i=1}^{n}(-\varepsilon_i) = 0 \tag{40}$$

$$\sum_{i=1}^{n}\varepsilon_i = 0 \tag{41}$$

$$\bar{\varepsilon} = 0 \tag{42}$$

$$\boxed{\mathbb{E}[\varepsilon] = 0}$$

Uncorrelated with Predictor

By definition:

$$\mathrm{Cov}(X,\varepsilon) = \mathbb{E}[(X - \mu_X)(\varepsilon - \mu_\varepsilon)] = \mathbb{E}[X\varepsilon] - \mathbb{E}[X]\mathbb{E}[\varepsilon]$$

Since $\mathbb{E}[\varepsilon] = 0$, this reduces to $\mathrm{Cov}(X,\varepsilon) = \mathbb{E}[X\varepsilon]$.

Substituting $\varepsilon = Y - a - bX$:

$$\mathbb{E}[X\varepsilon] = \mathbb{E}[X(Y - a - bX)] = \mathbb{E}[XY] - a\mathbb{E}[X] - b\mathbb{E}[X^2]$$

From the first-order condition for b:

$$-\frac{2}{n}\sum_{i=1}^{n}x_i(y_i - a - bx_i) = 0 \implies \sum_{i=1}^{n}x_i(y_i - a - bx_i) = 0$$

Dividing by n:

$$\frac{1}{n}\sum_{i=1}^{n}x_iy_i - a\frac{1}{n}\sum_{i=1}^{n}x_i - b\frac{1}{n}\sum_{i=1}^{n}x_i^2 = 0$$

This is the sample analogue of $\mathbb{E}[XY] - a\mathbb{E}[X] - b\mathbb{E}[X^2] = 0$. Therefore:

$$\mathrm{Cov}(X, \varepsilon) = \mathbb{E}[X\varepsilon] = \mathbb{E}[XY] - a\mathbb{E}[X] - b\mathbb{E}[X^2] = 0$$

$$\boxed{\mathrm{Cov}(X, \varepsilon) = 0}$$

Summary (Univariate Case)

For the linear model $Y = a + bX + \varepsilon$ with OLS estimators:

1. $\mathbb{E}[\varepsilon] = 0$ - residuals have zero mean
2. $\mathrm{Cov}(X, \varepsilon) = 0$ - residuals are uncorrelated with the predictor

These are not assumptions but consequences of the first-order conditions.

Matrix Notation

Let us now consider the matrix notation, which is helpful to generalize our model for higher dimensions. For each observation $i = 1, \ldots, n$, let $\mathbf{x}_i \in \mathbb{R}^d$ be the vector of explanatory variables and $y_i \in \mathbb{R}$ the dependent variable. The model is:

$$y_i = \mathbf{x}_i^\top \boldsymbol{\theta} + \varepsilon_i$$

where $\boldsymbol{\theta} \in \mathbb{R}^d$ is the parameter vector and $\{\varepsilon_i\}_{i=1,\ldots,n}$ are noise terms satisfying $\mathrm{cov}(\mathbf{x}_i, \varepsilon_i) = \mathbf{0}$.

To include an intercept, we assume the first coordinate of $\mathbf{x}_i$ is 1:

$$\mathbf{x}_i = \begin{bmatrix} 1 \\ x_{i1} \\ \vdots \\ x_{i,d-1} \end{bmatrix}, \quad \boldsymbol{\theta} = \begin{bmatrix} a \\ b_1 \\ \vdots \\ b_{d-1} \end{bmatrix}$$

so that $\mathbf{x}_i^\top \boldsymbol{\theta} = a + b_1 x_{i1} + \cdots + b_{d-1} x_{i,d-1}$, where $\theta_1 = a$ is the intercept.

Let $\mathbf{y} = (y_1, \ldots, y_n)^\top \in \mathbb{R}^n$ be the vector of responses and $\mathbf{X} \in \mathbb{R}^{n \times d}$ the matrix whose i-th row is $\mathbf{x}_i^\top$:

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix} 1 & x_{11} & \cdots & x_{1,d-1} \\ 1 & x_{21} & \cdots & x_{2,d-1} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \cdots & x_{n,d-1} \end{bmatrix}$$

The matrix $\mathbf{X}$ is called the *design matrix*. The model can be written compactly as:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\theta} + \boldsymbol{\varepsilon}$$

The goal is to find an estimator $\hat{\boldsymbol{\theta}}$ of the true parameter $\boldsymbol{\theta}$. The empirical risk is:

$$\hat{\mathcal{R}}_n(\boldsymbol{\theta}) = \frac{1}{n} \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|_2^2$$

Deriving the OLS Estimator (Matrix Form)

We expand the squared norm:

$$\hat{\mathcal{R}}_n(\boldsymbol{\theta}) = \frac{1}{n}(\mathbf{y} - \mathbf{X}\boldsymbol{\theta})^\top(\mathbf{y} - \mathbf{X}\boldsymbol{\theta}) \tag{43}$$

$$= \frac{1}{n}\left(\mathbf{y}^\top\mathbf{y} - \mathbf{y}^\top\mathbf{X}\boldsymbol{\theta} - \boldsymbol{\theta}^\top\mathbf{X}^\top\mathbf{y} + \boldsymbol{\theta}^\top\mathbf{X}^\top\mathbf{X}\boldsymbol{\theta}\right) \tag{44}$$

$$= \frac{1}{n}\left(\boldsymbol{\theta}^\top\mathbf{X}^\top\mathbf{X}\boldsymbol{\theta} - 2\boldsymbol{\theta}^\top\mathbf{X}^\top\mathbf{y} + \|\mathbf{y}\|^2\right) \tag{45}$$

The last line follows because $\mathbf{y}^\top\mathbf{X}\boldsymbol{\theta} = \boldsymbol{\theta}^\top\mathbf{X}^\top\mathbf{y}$, as the transpose of a scalar is itself.

Since $\hat{\mathcal{R}}_n$ is differentiable, any minimizer must satisfy $\nabla\hat{\mathcal{R}}_n(\hat{\boldsymbol{\theta}}) = \mathbf{0}$. We compute the gradient using standard matrix calculus identities: for symmetric $\mathbf{A}$, we have $\nabla_{\boldsymbol{\theta}}(\boldsymbol{\theta}^\top\mathbf{A}\boldsymbol{\theta}) = 2\mathbf{A}\boldsymbol{\theta}$ and $\nabla_{\boldsymbol{\theta}}(\boldsymbol{\theta}^\top\mathbf{b}) = \mathbf{b}$. Since $\mathbf{X}^\top\mathbf{X} \in \mathbb{R}^{d\times d}$ is symmetric:

$$\nabla\hat{\mathcal{R}}_n(\boldsymbol{\theta}) = \frac{1}{n}\left(2\mathbf{X}^\top\mathbf{X}\boldsymbol{\theta} - 2\mathbf{X}^\top\mathbf{y}\right) \tag{46}$$

$$= \frac{2}{n}\left(\mathbf{X}^\top\mathbf{X}\boldsymbol{\theta} - \mathbf{X}^\top\mathbf{y}\right) \tag{47}$$

Setting the gradient to zero yields the *normal equations*:

$$\mathbf{X}^\top\mathbf{X}\hat{\boldsymbol{\theta}} = \mathbf{X}^\top\mathbf{y}$$

If $\mathbf{X}^\top\mathbf{X}$ is invertible, we can multiply both sides by $(\mathbf{X}^\top\mathbf{X})^{-1}$:

$$\boxed{\hat{\boldsymbol{\theta}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y}}$$

The rank condition. The inverse $(\mathbf{X}^\top\mathbf{X})^{-1}$ exists iff $\operatorname{rank}(\mathbf{X}) = d$. Without full column rank, infinitely many $\boldsymbol{\theta}$ satisfy the normal equations. Since $\operatorname{rank}(\mathbf{X}) \leq \min(n, d)$, we need $n \geq d$. The case $\operatorname{rank}(\mathbf{X}) < d$ is called *perfect multicollinearity*.

Verifying the minimum. The Hessian is $\nabla^2\hat{\mathcal{R}}_n(\boldsymbol{\theta}) = \frac{2}{n}\mathbf{X}^\top\mathbf{X}$. For any $\mathbf{v} \neq \mathbf{0}$:

$$\mathbf{v}^\top(\mathbf{X}^\top\mathbf{X})\mathbf{v} = \|\mathbf{X}\mathbf{v}\|^2 > 0$$

The strict inequality holds because full column rank implies $\mathbf{X}\mathbf{v} \neq \mathbf{0}$ for $\mathbf{v} \neq \mathbf{0}$. The Hessian is positive definite, so $\hat{\mathcal{R}}_n$ is strictly convex and $\hat{\boldsymbol{\theta}}$ is the unique global minimum.

Geometric Interpretation

The residual vector $\mathbf{y} - \hat{\mathbf{y}} \in \mathbb{R}^n$ has coordinates:

$$\mathbf{y} - \hat{\mathbf{y}} = \begin{bmatrix} y_1 - \hat{y}_1 \\ y_2 - \hat{y}_2 \\ \vdots \\ y_n - \hat{y}_n \end{bmatrix}$$

Each entry is one vertical residual. The squared Euclidean norm of this vector is:

$$\|\mathbf{y} - \hat{\mathbf{y}}\|_2^2 = (y_1 - \hat{y}_1)^2 + (y_2 - \hat{y}_2)^2 + \cdots + (y_n - \hat{y}_n)^2 = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

So minimizing the Euclidean distance in $\mathbb{R}^n$ between $\mathbf{y}$ and $\hat{\mathbf{y}}$ is exactly minimizing the sum of squared vertical distances.

The vector $\mathbf{y} \in \mathbb{R}^n$ generally does not lie in $\mathrm{Im}(\mathbf{X}) = \{\mathbf{X}\boldsymbol{\theta} : \boldsymbol{\theta} \in \mathbb{R}^d\}$, the d-dimensional subspace spanned by the columns of $\mathbf{X}$. The closest point in this subspace is the orthogonal projection of $\mathbf{y}$ onto $\mathrm{Im}(\mathbf{X})$:

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\theta}} = \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y} = \mathbf{P}_\mathbf{X}\mathbf{y}$$

The matrix $\mathbf{P}_\mathbf{X} := \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top$ is the projection matrix onto $\mathrm{Im}(\mathbf{X})$. The residual vector $\hat{\boldsymbol{\varepsilon}} = \mathbf{y} - \hat{\mathbf{y}}$ is orthogonal to every column of $\mathbf{X}$, which is the content of the normal equations: $\mathbf{X}^\top(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\theta}}) = \mathbf{0}$.

Numerical Methods

The closed-form $\hat{\boldsymbol{\theta}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y}$ requires $O(nd^2)$ to form $\mathbf{X}^\top\mathbf{X}$ and $O(d^3)$ to invert it. When d is large, this becomes infeasible.

Gradient descent bypasses matrix inversion by iterating. Recall the gradient:

$$\nabla\hat{\mathcal{R}}_n(\boldsymbol{\theta}) = \frac{2}{n}\left(\mathbf{X}^\top\mathbf{X}\boldsymbol{\theta} - \mathbf{X}^\top\mathbf{y}\right)$$

Starting from $\hat{\boldsymbol{\theta}}_0 = \mathbf{0}$, we update:

$$\hat{\boldsymbol{\theta}}_{t+1} = \hat{\boldsymbol{\theta}}_t - \eta\nabla\hat{\mathcal{R}}_n(\hat{\boldsymbol{\theta}}_t) = \hat{\boldsymbol{\theta}}_t - \frac{2\eta}{n}\left(\mathbf{X}^\top\mathbf{X}\hat{\boldsymbol{\theta}}_t - \mathbf{X}^\top\mathbf{y}\right)$$

At convergence, $\hat{\boldsymbol{\theta}}_{t+1} = \hat{\boldsymbol{\theta}}_t$, which requires $\mathbf{X}^\top\mathbf{X}\hat{\boldsymbol{\theta}} = \mathbf{X}^\top\mathbf{y}$ - the normal equations. So gradient descent converges to the OLS solution.

When n is large, computing the full gradient requires summing over all observations:

$$\nabla\hat{\mathcal{R}}_n(\boldsymbol{\theta}) = \frac{2}{n}\sum_{i=1}^{n}\mathbf{x}_i(\mathbf{x}_i^\top\boldsymbol{\theta} - y_i)$$

Stochastic gradient descent (SGD) approximates this using a single observation:

$$\nabla_i(\boldsymbol{\theta}) = 2\mathbf{x}_i(\mathbf{x}_i^\top\boldsymbol{\theta} - y_i)$$

This is an unbiased estimate: $\mathbb{E}_i[\nabla_i(\boldsymbol{\theta})] = \nabla\hat{\mathcal{R}}_n(\boldsymbol{\theta})$. The SGD update is:

$$\hat{\boldsymbol{\theta}}_{t+1} = \hat{\boldsymbol{\theta}}_t - \eta_t\mathbf{x}_i(\mathbf{x}_i^\top\hat{\boldsymbol{\theta}}_t - y_i)$$

With a decaying learning rate $\eta_t \to 0$ (such that $\sum_t \eta_t = \infty$ and $\sum_t \eta_t^2 < \infty$), SGD converges to the OLS solution. Mini-batch gradient descent averages over a subset of B observations, reducing variance while remaining tractable.

The Classical Linear Model Assumptions

So far, we have treated regression purely as an optimization problem. To make *statistical* statements about $\hat{\boldsymbol{\theta}}$-its distribution, confidence intervals, hypothesis tests-we need assumptions about the data-generating process.

Definition 1 (Classical Linear Model)**.** *The **fixed-X** or **classical linear model** assumes:*

1. ***Linearity**: $\mathbf{y} = \mathbf{X}\boldsymbol{\theta} + \boldsymbol{\varepsilon}$ for some true $\boldsymbol{\theta} \in \mathbb{R}^d$*
2. ***Fixed design**: $\mathbf{X}$ is non-random (fixed across repeated samples)*
3. ***Strict exogeneity**: $\mathbb{E}[\boldsymbol{\varepsilon}|\mathbf{X}] = \mathbf{0}$*
4. ***Spherical errors**: $Var(\boldsymbol{\varepsilon}|\mathbf{X}) = \sigma^2\mathbf{I}_n$ (homoskedasticity + no autocorrelation)*
5. ***Full rank**: $rank(\mathbf{X}) = d < n$*

Assumption (3) implies $\mathbb{E}[\varepsilon_i] = 0$ and $\mathrm{Cov}(\mathbf{x}_i, \varepsilon_i) = \mathbf{0}$. Assumption (4) says errors have constant variance σ^2 and are uncorrelated with each other.

Remark 2. *For inference (confidence intervals, t-tests), we add:*

6. ***Normality**: $\boldsymbol{\varepsilon}|\mathbf{X} \sim \mathcal{N}(\mathbf{0}, \sigma^2\mathbf{I}_n)$*

Under (1)–(5) alone, we get unbiasedness and BLUE. Adding (6) gives exact finite-sample distributions.

Unbiasedness of OLS

Theorem 3 (Unbiasedness)**.** *Under assumptions (1)–(3), the OLS estimator is unbiased: $\mathbb{E}[\hat{\boldsymbol{\theta}}] = \boldsymbol{\theta}$.*

Proof. Substitute the true model $\mathbf{y} = \mathbf{X}\boldsymbol{\theta} + \boldsymbol{\varepsilon}$ into the OLS formula:

$$\begin{aligned}
\hat{\boldsymbol{\theta}} &= (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y} && (48)\\
&= (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top(\mathbf{X}\boldsymbol{\theta} + \boldsymbol{\varepsilon}) && (49)\\
&= (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{X}\boldsymbol{\theta} + (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\boldsymbol{\varepsilon} && (50)\\
&= \boldsymbol{\theta} + (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\boldsymbol{\varepsilon} && (51)
\end{aligned}$$

Taking expectations (with $\mathbf{X}$ fixed):

$$\mathbb{E}[\hat{\boldsymbol{\theta}}] = \boldsymbol{\theta} + (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbb{E}[\boldsymbol{\varepsilon}] = \boldsymbol{\theta} + \mathbf{0} = \boldsymbol{\theta}$$

□

Variance of the OLS Estimator

Theorem 4 (Variance of OLS)**.** *Under assumptions (1)–(4):*

$$Var(\hat{\boldsymbol{\theta}}) = \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}$$

Proof. From $\hat{\boldsymbol{\theta}} = \boldsymbol{\theta} + (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\boldsymbol{\varepsilon}$:

$$\begin{aligned}
\mathrm{Var}(\hat{\boldsymbol{\theta}}) &= \mathrm{Var}\left((\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\boldsymbol{\varepsilon}\right) && (52)\\
&= (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathrm{Var}(\boldsymbol{\varepsilon})\mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1} && (53)\\
&= (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top(\sigma^2\mathbf{I}_n)\mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1} && (54)\\
&= \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1} && (55)\\
&= \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1} && (56)
\end{aligned}$$

□

The variance of the j-th coefficient is $\mathrm{Var}(\hat{\theta}_j) = \sigma^2[(\mathbf{X}^\top\mathbf{X})^{-1}]_{jj}$.

The Gauss-Markov Theorem

The Gauss-Markov theorem establishes OLS as the *best* among a restricted class of estimators.

Definition 5 (Linear Estimator)**.** *An estimator $\tilde{\boldsymbol{\theta}}$ is **linear** in $\mathbf{y}$ if $\tilde{\boldsymbol{\theta}} = \mathbf{A}\mathbf{y}$ for some matrix $\mathbf{A} \in \mathbb{R}^{d\times n}$ that depends only on $\mathbf{X}$.*

OLS is linear: $\hat{\boldsymbol{\theta}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y} = \mathbf{A}_{\mathrm{OLS}}\mathbf{y}$ with $\mathbf{A}_{\mathrm{OLS}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top$.

Theorem 6 (Gauss-Markov)**.** *Under assumptions (1)–(5), the OLS estimator is **BLUE**: Best Linear Unbiased Estimator. That is, for any linear unbiased estimator $\tilde{\boldsymbol{\theta}} = \mathbf{A}\mathbf{y}$:*

$$Var(\tilde{\theta}_j) \geq Var(\hat{\theta}_j) \quad \text{for all } j = 1, \ldots, d$$

Equivalently, $Var(\tilde{\boldsymbol{\theta}}) - Var(\hat{\boldsymbol{\theta}})$ is positive semi-definite.

Proof. Let $\tilde{\boldsymbol{\theta}} = \mathbf{A}\mathbf{y}$ be any linear unbiased estimator. Write $\mathbf{A} = \mathbf{A}_{\mathrm{OLS}} + \mathbf{D}$ where $\mathbf{D} = \mathbf{A} - \mathbf{A}_{\mathrm{OLS}}$.

Step 1: Unbiasedness constraint. For $\tilde{\boldsymbol{\theta}}$ to be unbiased:

$$\mathbb{E}[\tilde{\boldsymbol{\theta}}] = \mathbb{E}[\mathbf{A}(\mathbf{X}\boldsymbol{\theta} + \boldsymbol{\varepsilon})] = \mathbf{A}\mathbf{X}\boldsymbol{\theta} = \boldsymbol{\theta} \quad \forall\boldsymbol{\theta} \tag{57}$$

This requires $\mathbf{A}\mathbf{X} = \mathbf{I}_d$. Since $\mathbf{A}_{\mathrm{OLS}}\mathbf{X} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{X} = \mathbf{I}_d$, we need:

$$\mathbf{D}\mathbf{X} = (\mathbf{A} - \mathbf{A}_{\mathrm{OLS}})\mathbf{X} = \mathbf{I}_d - \mathbf{I}_d = \mathbf{0}$$

Step 2: Variance of $\tilde{\boldsymbol{\theta}}$.

$$\begin{aligned}
\mathrm{Var}(\tilde{\boldsymbol{\theta}}) &= \mathrm{Var}(\mathbf{A}\boldsymbol{\varepsilon}) = \sigma^2\mathbf{A}\mathbf{A}^\top && (58)\\
&= \sigma^2(\mathbf{A}_{\mathrm{OLS}} + \mathbf{D})(\mathbf{A}_{\mathrm{OLS}} + \mathbf{D})^\top && (59)\\
&= \sigma^2\left(\mathbf{A}_{\mathrm{OLS}}\mathbf{A}_{\mathrm{OLS}}^\top + \mathbf{A}_{\mathrm{OLS}}\mathbf{D}^\top + \mathbf{D}\mathbf{A}_{\mathrm{OLS}}^\top + \mathbf{D}\mathbf{D}^\top\right) && (60)
\end{aligned}$$

Step 3: Cross terms vanish. Note that $\mathbf{A}_{\text{OLS}}^\top = \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}$. Since $\mathbf{DX} = \mathbf{0}$:

$$\mathbf{DA}_{\text{OLS}}^\top = \mathbf{DX}(\mathbf{X}^\top\mathbf{X})^{-1} = \mathbf{0}$$

Similarly, $\mathbf{A}_{\text{OLS}}\mathbf{D}^\top = (\mathbf{DA}_{\text{OLS}}^\top)^\top = \mathbf{0}$.

Step 4: Conclusion.

$$\begin{aligned}\text{Var}(\tilde{\boldsymbol{\theta}}) &= \sigma^2\mathbf{A}_{\text{OLS}}\mathbf{A}_{\text{OLS}}^\top + \sigma^2\mathbf{DD}^\top && (61)\\ &= \text{Var}(\hat{\boldsymbol{\theta}}) + \sigma^2\mathbf{DD}^\top && (62)\end{aligned}$$

Since $\mathbf{DD}^\top$ is positive semi-definite, $\text{Var}(\tilde{\boldsymbol{\theta}}) \geq \text{Var}(\hat{\boldsymbol{\theta}})$. □

Remark 7. *BLUE means best among linear unbiased estimators. Nonlinear or biased estimators may have lower MSE (see Part III on regularization).*

Beyond BLUE: The BUE Condition

Under normality (assumption 6), we can strengthen Gauss-Markov.

Theorem 8 (OLS is BUE under Normality)**.** *Under assumptions (1)–(6), OLS is **BUE**: Best Unbiased Estimator (not just linear). It achieves the Cramér-Rao lower bound.*

The Cramér-Rao bound states that for any unbiased estimator $\tilde{\boldsymbol{\theta}}$:

$$\text{Var}(\tilde{\boldsymbol{\theta}}) \geq \mathcal{I}(\boldsymbol{\theta})^{-1}$$

where $\mathcal{I}(\boldsymbol{\theta})$ is the Fisher information matrix. Under the normal linear model, $\mathcal{I}(\boldsymbol{\theta}) = \frac{1}{\sigma^2}\mathbf{X}^\top\mathbf{X}$, so the bound is $\sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}$-exactly the variance of OLS.

Estimating σ^2

The error variance σ^2 is unknown and must be estimated. A natural estimator uses the residuals:

$$\hat{\sigma}^2 = \frac{1}{n-d}\sum_{i=1}^{n}\hat{\varepsilon}_i^2 = \frac{\|\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\theta}}\|^2}{n-d} = \frac{\text{RSS}}{n-d}$$

where RSS = Residual Sum of Squares.

Theorem 9. *Under assumptions (1)–(5), $\hat{\sigma}^2$ is unbiased: $\mathbb{E}[\hat{\sigma}^2] = \sigma^2$.*

Proof. The residual vector is $\hat{\boldsymbol{\varepsilon}} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\theta}} = (\mathbf{I} - \mathbf{P}_\mathbf{X})\mathbf{y} = \mathbf{M}\boldsymbol{\varepsilon}$, where $\mathbf{M} = \mathbf{I} - \mathbf{P}_\mathbf{X}$.

The matrix $\mathbf{M}$ is symmetric and idempotent ($\mathbf{M}^2 = \mathbf{M}$), with $\text{tr}(\mathbf{M}) = n - d$.

$$\mathbb{E}[\text{RSS}] = \mathbb{E}[\hat{\boldsymbol{\varepsilon}}^\top\hat{\boldsymbol{\varepsilon}}] = \mathbb{E}[\boldsymbol{\varepsilon}^\top\mathbf{M}^\top\mathbf{M}\boldsymbol{\varepsilon}] = \mathbb{E}[\boldsymbol{\varepsilon}^\top\mathbf{M}\boldsymbol{\varepsilon}] \tag{63}$$

Using $\mathbb{E}[\boldsymbol{\varepsilon}^\top\mathbf{A}\boldsymbol{\varepsilon}] = \text{tr}(\mathbf{A}\text{Var}(\boldsymbol{\varepsilon})) + \mathbb{E}[\boldsymbol{\varepsilon}]^\top\mathbf{A}\mathbb{E}[\boldsymbol{\varepsilon}]$ with $\mathbb{E}[\boldsymbol{\varepsilon}] = \mathbf{0}$:

$$\mathbb{E}[\text{RSS}] = \text{tr}(\mathbf{M}\cdot\sigma^2\mathbf{I}) = \sigma^2\text{tr}(\mathbf{M}) = \sigma^2(n-d)$$

Hence $\mathbb{E}[\hat{\sigma}^2] = \sigma^2$. □

Sampling Distribution of $\hat{\boldsymbol{\theta}}$

Under normality (assumption 6):

Theorem 10. *Under assumptions (1)–(6):*

$$\hat{\boldsymbol{\theta}} \sim \mathcal{N}\left(\boldsymbol{\theta}, \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}\right)$$

This follows because $\hat{\boldsymbol{\theta}} = \boldsymbol{\theta} + (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\boldsymbol{\varepsilon}$ is a linear transformation of the normal vector $\boldsymbol{\varepsilon}$.

For individual coefficients:

$$\hat{\theta}_j \sim \mathcal{N}\left(\theta_j, \sigma^2[(\mathbf{X}^\top\mathbf{X})^{-1}]_{jj}\right)$$

The t-Test for Coefficients

Since σ^2 is unknown, we use $\hat{\sigma}^2$ to form a t-statistic.

Theorem 11. *Under assumptions (1)–(6):*

$$t_j = \frac{\hat{\theta}_j - \theta_j}{se(\hat{\theta}_j)} \sim t_{n-d}$$

where $se(\hat{\theta}_j) = \hat{\sigma}\sqrt{[(\mathbf{X}^\top\mathbf{X})^{-1}]_{jj}}$ *is the standard error.*

To test $H_0 : \theta_j = 0$ vs $H_1 : \theta_j \neq 0$, compute $t_j = \hat{\theta}_j/\mathrm{se}(\hat{\theta}_j)$ and reject if $|t_j| > t_{n-d,\alpha/2}$.

A $(1-\alpha)$ confidence interval for θ_j:

$$\hat{\theta}_j \pm t_{n-d,\alpha/2} \cdot \mathrm{se}(\hat{\theta}_j)$$

The F-Test for Multiple Coefficients

To test whether a subset of coefficients are jointly zero, use the F-test.

Setup: Partition $\boldsymbol{\theta} = (\boldsymbol{\theta}_1^\top, \boldsymbol{\theta}_2^\top)^\top$ where $\boldsymbol{\theta}_1 \in \mathbb{R}^{d_1}$ (intercept and retained variables) and $\boldsymbol{\theta}_2 \in \mathbb{R}^{d_2}$ (variables to test). We test $H_0 : \boldsymbol{\theta}_2 = \mathbf{0}$.

Let $\mathrm{RSS}_{\mathrm{full}}$ = RSS from the full model, $\mathrm{RSS}_{\mathrm{reduced}}$ = RSS from the model without $\boldsymbol{\theta}_2$.

Theorem 12. *Under H_0 and assumptions (1)–(6):*

$$F = \frac{(RSS_{reduced} - RSS_{full})/d_2}{RSS_{full}/(n-d)} \sim F_{d_2,n-d}$$

Reject H_0 if $F > F_{d_2,n-d,\alpha}$.

Mean Squared Error Decomposition

The MSE of an estimator $\hat{\theta}$ for a parameter θ is:

$$\text{MSE}(\hat{\theta}) = \mathbb{E}[(\hat{\theta} - \theta)^2]$$

Theorem 13 (Bias-Variance Decomposition)**.**

$$MSE(\hat{\theta}) = Bias(\hat{\theta})^2 + Var(\hat{\theta})$$

where $Bias(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta$.

Proof. Let $\mu = \mathbb{E}[\hat{\theta}]$. Then:

$$\begin{aligned}
\text{MSE}(\hat{\theta}) &= \mathbb{E}[(\hat{\theta} - \theta)^2] && (64)\\
&= \mathbb{E}[(\hat{\theta} - \mu + \mu - \theta)^2] && (65)\\
&= \mathbb{E}[(\hat{\theta} - \mu)^2] + 2\mathbb{E}[(\hat{\theta} - \mu)(\mu - \theta)] + (\mu - \theta)^2 && (66)\\
&= \text{Var}(\hat{\theta}) + 2(\mu - \theta)\underbrace{\mathbb{E}[\hat{\theta} - \mu]}_{=0} + \text{Bias}(\hat{\theta})^2 && (67)\\
&= \text{Var}(\hat{\theta}) + \text{Bias}(\hat{\theta})^2 && (68)
\end{aligned}$$

□

Prediction Error Decomposition

For prediction at a new point $\mathbf{x}_0$ with true value $y_0 = \mathbf{x}_0^\top \boldsymbol{\theta} + \varepsilon_0$:

Theorem 14 (Prediction MSE Decomposition)**.**

$$\mathbb{E}[(y_0 - \hat{y}_0)^2] = \sigma^2 + Bias(\hat{y}_0)^2 + Var(\hat{y}_0)$$

where $\hat{y}_0 = \mathbf{x}_0^\top \hat{\boldsymbol{\theta}}$.

Proof. Let $f_0 = \mathbf{x}_0^\top \boldsymbol{\theta}$ (true signal) and $\hat{f}_0 = \hat{y}_0$ (prediction). Note $y_0 = f_0 + \varepsilon_0$ with ε_0 independent of $\hat{f}_0$.

$$\begin{aligned}
\mathbb{E}[(y_0 - \hat{f}_0)^2] &= \mathbb{E}[(f_0 + \varepsilon_0 - \hat{f}_0)^2] && (69)\\
&= \mathbb{E}[(f_0 - \hat{f}_0)^2] + \mathbb{E}[\varepsilon_0^2] + 2\mathbb{E}[(f_0 - \hat{f}_0)\varepsilon_0] && (70)\\
&= \mathbb{E}[(f_0 - \hat{f}_0)^2] + \sigma^2 + 0 && (71)\\
&= \text{MSE}(\hat{f}_0) + \sigma^2 && (72)\\
&= \text{Bias}(\hat{f}_0)^2 + \text{Var}(\hat{f}_0) + \sigma^2 && (73)
\end{aligned}$$

□

The three terms are:

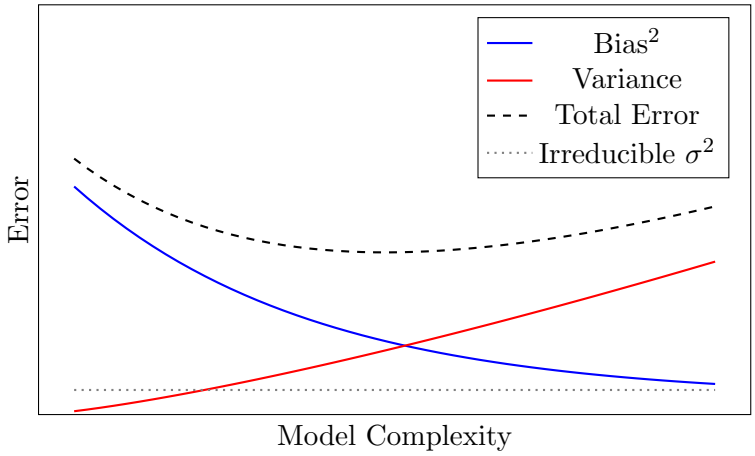
- σ^2: **Irreducible error** - noise in the data
- $\text{Bias}(\hat{f}_0)^2$: **Bias squared** - systematic error from model misspecification
- $\text{Var}(\hat{f}_0)$: **Variance** - sensitivity to training data

The Tradeoff in Practice

High bias (underfitting): Model is too simple. Cannot capture the true relationship. Example: fitting a line to quadratic data.

High variance (overfitting): Model is too complex. Fits noise in the training data, performs poorly on new data. Example: high-degree polynomial with few data points.

Bias-Variance Tradeoff



The optimal model complexity minimizes total error-balancing bias and variance.

Detecting Overfitting and Underfitting

Training Error vs Test Error

Overfitting signature: Low training error, high test error. The gap between them indicates overfitting.

Underfitting signature: Both training and test error are high. The model cannot fit even the training data well.

Cross-Validation

k-fold cross-validation estimates out-of-sample error:

1. Split data into k folds
2. For each fold i: train on all data except fold i, evaluate on fold i
3. Average the k test errors

The CV estimate:

$$\text{CV}_k = \frac{1}{k}\sum_{i=1}^{k} \text{MSE}_i$$

Leave-one-out CV ($k = n$) has low bias but high variance. $k = 5$ or $k = 10$ is typical.

Learning Curves

Plot training and test error as a function of training set size n:

- **High bias:** Both curves plateau at high error
- **High variance:** Large gap between curves; test error decreases slowly with n

Ridge Regression (Tikhonov / L_2 Regularization)

Regularization introduces bias to reduce variance, often improving out-of-sample prediction. The key insight: a biased estimator can have lower MSE than an unbiased one if the variance reduction outweighs the bias introduced.

Motivation

When $\mathbf{X}^\top\mathbf{X}$ is nearly singular (multicollinearity) or $d > n$, OLS has high variance. Ridge regression stabilizes the solution by adding a penalty on $\|\boldsymbol{\theta}\|_2^2$.

The Ridge Objective

Definition 15 (Ridge Regression)**.** *The ridge estimator minimizes:*

$$\hat{\boldsymbol{\theta}}_{ridge} = \arg\min_{\boldsymbol{\theta}} \left\{ \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|_2^2 + \lambda\|\boldsymbol{\theta}\|_2^2 \right\}$$

where $\lambda > 0$ is the regularization parameter.

Closed-Form Solution

Theorem 16. *The ridge estimator is:*

$$\boxed{\hat{\boldsymbol{\theta}}_{ridge} = (\mathbf{X}^\top\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}^\top\mathbf{y}}$$

Proof. The objective is $L(\boldsymbol{\theta}) = (\mathbf{y} - \mathbf{X}\boldsymbol{\theta})^\top(\mathbf{y} - \mathbf{X}\boldsymbol{\theta}) + \lambda\boldsymbol{\theta}^\top\boldsymbol{\theta}$.

Expanding:

$$L(\boldsymbol{\theta}) = \mathbf{y}^\top\mathbf{y} - 2\boldsymbol{\theta}^\top\mathbf{X}^\top\mathbf{y} + \boldsymbol{\theta}^\top\mathbf{X}^\top\mathbf{X}\boldsymbol{\theta} + \lambda\boldsymbol{\theta}^\top\boldsymbol{\theta}$$

Taking the gradient:

$$\nabla L = -2\mathbf{X}^\top\mathbf{y} + 2\mathbf{X}^\top\mathbf{X}\boldsymbol{\theta} + 2\lambda\boldsymbol{\theta} = 2\left[(\mathbf{X}^\top\mathbf{X} + \lambda\mathbf{I})\boldsymbol{\theta} - \mathbf{X}^\top\mathbf{y}\right]$$

Setting to zero:

$$(\mathbf{X}^\top\mathbf{X} + \lambda\mathbf{I})\hat{\boldsymbol{\theta}}_{\text{ridge}} = \mathbf{X}^\top\mathbf{y}$$

Since $\mathbf{X}^\top\mathbf{X} + \lambda\mathbf{I}$ is always invertible for $\lambda > 0$ (eigenvalues are shifted by λ):

$$\hat{\boldsymbol{\theta}}_{\text{ridge}} = (\mathbf{X}^\top\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}^\top\mathbf{y}$$

□

Bayesian Interpretation

Ridge regression is the posterior mode under:

- Prior: $\boldsymbol{\theta} \sim \mathcal{N}(\mathbf{0}, \tau^2\mathbf{I})$
- Likelihood: $\mathbf{y}|\boldsymbol{\theta} \sim \mathcal{N}(\mathbf{X}\boldsymbol{\theta}, \sigma^2\mathbf{I})$

with $\lambda = \sigma^2/\tau^2$. A larger λ corresponds to a stronger prior belief that $\boldsymbol{\theta}$ is near zero.

Bias and Variance of Ridge

Theorem 17. *The ridge estimator has:*

$$\mathbb{E}[\hat{\boldsymbol{\theta}}_{ridge}] = (\mathbf{X}^\top\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}^\top\mathbf{X}\boldsymbol{\theta} = \mathbf{W}\boldsymbol{\theta} \neq \boldsymbol{\theta} \tag{74}$$

$$Var(\hat{\boldsymbol{\theta}}_{ridge}) = \sigma^2\mathbf{W}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{W}^\top \tag{75}$$

where $\mathbf{W} = (\mathbf{X}^\top\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}^\top\mathbf{X}$.

Key insight: $\mathbf{W} \neq \mathbf{I}$, so ridge is *biased*. But $\mathrm{Var}(\hat{\boldsymbol{\theta}}_{\text{ridge}}) < \mathrm{Var}(\hat{\boldsymbol{\theta}}_{\text{OLS}})$ for $\lambda > 0$, which can yield lower MSE.

SVD Interpretation

Let $\mathbf{X} = \mathbf{U}\mathbf{D}\mathbf{V}^\top$ be the SVD with singular values $d_1, \ldots, d_p$. Then:

$$\hat{\boldsymbol{\theta}}_{\text{ridge}} = \sum_{j=1}^{p} \frac{d_j^2}{d_j^2 + \lambda} \frac{\mathbf{u}_j^\top \mathbf{y}}{d_j} \mathbf{v}_j$$

Compare to OLS:

$$\hat{\boldsymbol{\theta}}_{\text{OLS}} = \sum_{j=1}^{p} \frac{\mathbf{u}_j^\top \mathbf{y}}{d_j} \mathbf{v}_j$$

Ridge shrinks coefficients corresponding to small singular values (high-variance directions) more aggressively via the factor $\frac{d_j^2}{d_j^2+\lambda} < 1$.

Lasso (L_1 Regularization)

The Lasso Objective

Definition 18 (Lasso)**.** *The Lasso estimator minimizes:*

$$\hat{\boldsymbol{\theta}}_{lasso} = \arg\min_{\boldsymbol{\theta}} \left\{ \frac{1}{2n} \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|_2^2 + \lambda \|\boldsymbol{\theta}\|_1 \right\}$$

where $\|\boldsymbol{\theta}\|_1 = \sum_{j=1}^{d} |\theta_j|$.

No Closed-Form Solution

Unlike ridge, Lasso has no closed-form solution due to the non-differentiability of $|\cdot|$ at zero. However, we can solve it via:

- Coordinate descent
- LARS (Least Angle Regression)
- Proximal gradient methods

Univariate Case: Soft Thresholding

For the simple problem $\min_\theta \frac{1}{2}(y-\theta)^2 + \lambda|\theta|$, the solution is the *soft-thresholding operator*:

$$\hat{\theta} = S_\lambda(y) = \mathrm{sign}(y)(|y| - \lambda)_+ = \begin{cases} y - \lambda & \text{if } y > \lambda \\ 0 & \text{if } |y| \leq \lambda \\ y + \lambda & \text{if } y < -\lambda \end{cases}$$

Proof. The subgradient of $|\theta|$ at θ is:

$$\partial|\theta| = \begin{cases} \{1\} & \text{if } \theta > 0 \\ [-1, 1] & \text{if } \theta = 0 \\ \{-1\} & \text{if } \theta < 0 \end{cases}$$

Setting $0 \in \partial\left[\frac{1}{2}(y-\theta)^2 + \lambda|\theta|\right]$:

$$0 = -(y - \theta) + \lambda \cdot \partial|\theta|$$

Case 1: $\hat{\theta} > 0$. Then $y - \hat{\theta} = \lambda$, so $\hat{\theta} = y - \lambda$. Valid when $y > \lambda$.

Case 2: $\hat{\theta} < 0$. Then $y - \hat{\theta} = -\lambda$, so $\hat{\theta} = y + \lambda$. Valid when $y < -\lambda$.

Case 3: $\hat{\theta} = 0$. Need $y - 0 \in \lambda[-1, 1]$, i.e., $|y| \leq \lambda$. □

Sparsity: Why Lasso Yields Exact Zeros

The L_1 penalty has corners at zero, creating a non-smooth optimization landscape. Geometrically, the constraint region $\{\boldsymbol{\theta} : \|\boldsymbol{\theta}\|_1 \leq t\}$ is a diamond (cross-polytope) with vertices on the axes. The OLS solution's level curves are ellipses. Their first contact point often occurs at a vertex, setting some $\theta_j = 0$ exactly.

In contrast, the L_2 constraint region is a sphere with no corners, so first contact typically occurs at a point with all $\theta_j \neq 0$.

Lasso as Variable Selection

Lasso performs *automatic variable selection*: irrelevant features are driven to exactly zero. This is useful when d is large and we believe only a few features matter (sparsity assumption).

Elastic Net

Elastic Net combines L_1 and L_2 penalties:

$$\hat{\boldsymbol{\theta}}_{\text{EN}} = \arg\min_{\boldsymbol{\theta}} \left\{ \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|_2^2 + \lambda_1\|\boldsymbol{\theta}\|_1 + \lambda_2\|\boldsymbol{\theta}\|_2^2 \right\}$$

Benefits:

- Sparsity from L_1 (variable selection)
- Grouping effect from L_2 (correlated features selected together)
- Stability when $d > n$ (ridge component ensures unique solution)

Tikhonov Regularization: General Form

Ridge regression is a special case of **Tikhonov regularization**:

$$\hat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}} \left\{ \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|_2^2 + \|\boldsymbol{\Gamma}\boldsymbol{\theta}\|_2^2 \right\}$$

where $\boldsymbol{\Gamma} \in \mathbb{R}^{k \times d}$ is the *Tikhonov matrix*.

Special cases:

- $\boldsymbol{\Gamma} = \sqrt{\lambda}\mathbf{I}$: Standard ridge regression
- $\boldsymbol{\Gamma} = \sqrt{\lambda}\mathbf{D}_1$: First-order differences (penalizes non-smoothness)
- $\boldsymbol{\Gamma} = \sqrt{\lambda}\mathbf{D}_2$: Second-order differences (penalizes curvature)

Theorem 19. *The Tikhonov solution is:*

$$\hat{\boldsymbol{\theta}} = (\mathbf{X}^\top\mathbf{X} + \boldsymbol{\Gamma}^\top\boldsymbol{\Gamma})^{-1}\mathbf{X}^\top\mathbf{y}$$

Choosing the Regularization Parameter

Cross-Validation

Select λ minimizing CV error:

$$\lambda^* = \arg\min_{\lambda} \mathrm{CV}(\lambda)$$

Typically use a grid of λ values on a log scale.

Information Criteria

For ridge/Lasso, we can use generalized degrees of freedom. For ridge:

$$\mathrm{df}(\lambda) = \mathrm{tr}\left(\mathbf{X}(\mathbf{X}^\top\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}^\top\right) = \sum_{j=1}^{p} \frac{d_j^2}{d_j^2 + \lambda}$$

Then use AIC or BIC (see Section 17).

Coefficient of Determination (R^2)

Definition

The **coefficient of determination** measures the proportion of variance explained:

Definition 20.

$$R^2 = 1 - \frac{RSS}{TSS} = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

where TSS = Total Sum of Squares $= \sum (y_i - \bar{y})^2$.

Interpretation

- $R^2 = 1$: Perfect fit (RSS = 0)
- $R^2 = 0$: Model no better than predicting $\bar{y}$
- $R^2 \in [0, 1]$ for models with an intercept

Alternative Formula

For OLS with intercept:

$$R^2 = \frac{\text{ESS}}{\text{TSS}} = \frac{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

where ESS = Explained Sum of Squares. The identity TSS = ESS + RSS holds for OLS with intercept.

R^2 as Squared Correlation

Theorem 21. *For OLS with intercept:*

$$R^2 = Corr(y, \hat{y})^2$$

The Problem: R^2 Always Increases with More Features

Theorem 22. *Adding a feature never decreases R^2, even if the feature is pure noise.*

Proof. Let R_k^2 be the R^2 with k features. The model with $k+1$ features includes all models with k features (by setting the new coefficient to zero). Since OLS minimizes RSS, $\text{RSS}_{k+1} \leq \text{RSS}_k$, so:

$$R_{k+1}^2 = 1 - \frac{\text{RSS}_{k+1}}{\text{TSS}} \geq 1 - \frac{\text{RSS}_k}{\text{TSS}} = R_k^2$$

□

This makes R^2 unsuitable for model comparison-it favors overly complex models.

Adjusted R^2

Definition 23 (Adjusted R^2)**.**

$$\bar{R}^2 = 1 - \frac{RSS/(n-d)}{TSS/(n-1)} = 1 - \frac{n-1}{n-d}(1 - R^2)$$

Adjusted R^2 penalizes for the number of parameters d. Adding a useless feature increases d without proportionally decreasing RSS, so $\bar{R}^2$ may decrease.

In-Sample vs Out-of-Sample R^2

In-Sample (Training) R^2

The R^2 computed on training data is *optimistically biased*-the model was fit to this data.

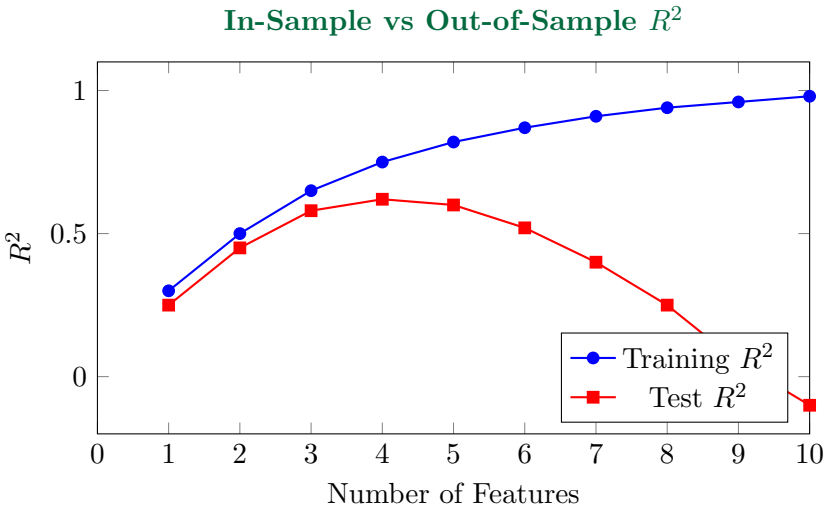
Out-of-Sample (Test) R^2

On new data $(x_i^*, y_i^*)_{i=1}^m$:

$$R_{\text{test}}^2 = 1 - \frac{\sum_{i=1}^m (y_i^* - \hat{y}_i^*)^2}{\sum_{i=1}^m (y_i^* - \bar{y}^*)^2}$$

This can be *negative* if the model performs worse than predicting the test mean.

Behavior with Model Complexity



Information Criteria

Information criteria balance fit (likelihood) against complexity (number of parameters).

Akaike Information Criterion (AIC)

Definition 24 (AIC)**.**

$$AIC = -2\log L(\hat{\boldsymbol{\theta}}) + 2d$$

where $L(\hat{\boldsymbol{\theta}})$ is the maximized likelihood and d is the number of parameters.

For linear regression with normal errors:

$$\text{AIC} = n\log\left(\frac{\text{RSS}}{n}\right) + 2d + \text{const}$$

Lower AIC is better. Select the model minimizing AIC.

Bayesian Information Criterion (BIC)

Definition 25 (BIC)**.**

$$BIC = -2\log L(\hat{\boldsymbol{\theta}}) + d\log n$$

BIC penalizes complexity more heavily than AIC when $n > e^2 \approx 7.4$. BIC is consistent (selects the true model as $n \to \infty$), while AIC tends to overfit.

Comparison

Criterion	Penalty	Property
AIC	$2d$	Efficient (minimizes prediction error)
BIC	$d\log n$	Consistent (finds true model)

Mallows' C_p and SURE

Mallows' C_p

Definition 26 (Mallows' C_p)**.**

$$C_p = \frac{RSS}{\hat{\sigma}^2} - n + 2d$$

where $\hat{\sigma}^2$ is estimated from the full model.

For a correctly specified model, $\mathbb{E}[C_p] \approx d$. Select models where $C_p \approx d$.

Stein's Unbiased Risk Estimate (SURE)

SURE provides an unbiased estimate of the true prediction risk without a test set.

Theorem 27 (SURE)**.** *For the model $\mathbf{y} = \boldsymbol{\mu} + \boldsymbol{\varepsilon}$ with $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2\mathbf{I})$, and estimator $\hat{\boldsymbol{\mu}} = g(\mathbf{y})$ with g weakly differentiable:*

$$SURE = -n\sigma^2 + \|\mathbf{y} - \hat{\boldsymbol{\mu}}\|^2 + 2\sigma^2 \sum_{i=1}^{n} \frac{\partial \hat{\mu}_i}{\partial y_i}$$

satisfies $\mathbb{E}[SURE] = \mathbb{E}[\|\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}\|^2]$.

For linear estimators $\hat{\boldsymbol{\mu}} = \mathbf{H}\mathbf{y}$, the divergence term is $\mathrm{tr}(\mathbf{H})$, the degrees of freedom.

Feature Selection Methods

Best Subset Selection

Enumerate all 2^d subsets, select the one minimizing CV error or information criterion. Computationally infeasible for $d > 20$.

Forward Stepwise Selection

1. Start with intercept only
2. Add the feature that most improves the criterion (CV, AIC, etc.)
3. Repeat until no improvement

Greedy, $O(d^2)$ model fits.

Backward Stepwise Selection

1. Start with all features
2. Remove the feature whose removal least hurts the criterion
3. Repeat until no improvement

Requires $n > d$.

Regularization Path

For Lasso, vary λ from large (all zeros) to small (OLS). The sequence of models entered gives a natural ordering of feature importance.

Kernel Methods

The Kernel Trick

Many learning algorithms depend on data only through inner products $\mathbf{x}_i^\top \mathbf{x}_j$. The *kernel trick* replaces these with $k(\mathbf{x}_i, \mathbf{x}_j)$ for a kernel function k, implicitly mapping to a high-dimensional feature space.

Definition 28 (Kernel)**.** *A function $k : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ is a **kernel** if it is symmetric and positive semi-definite: for any $\{x_1, \ldots, x_n\} \subset \mathcal{X}$, the matrix $\mathbf{K}$ with $K_{ij} = k(x_i, x_j)$ is positive semi-definite.*

By Mercer's theorem, such k corresponds to an inner product in some feature space: $k(\mathbf{x}, \mathbf{x}') = \langle \phi(\mathbf{x}), \phi(\mathbf{x}') \rangle$ for some $\phi : \mathcal{X} \to \mathcal{H}$.

Common Kernels

- **Linear**: $k(\mathbf{x}, \mathbf{x}') = \mathbf{x}^\top \mathbf{x}'$
- **Polynomial**: $k(\mathbf{x}, \mathbf{x}') = (1 + \mathbf{x}^\top \mathbf{x}')^p$
- **RBF/Gaussian**: $k(\mathbf{x}, \mathbf{x}') = \exp\left(-\frac{\|\mathbf{x}-\mathbf{x}'\|^2}{2\sigma^2}\right)$

Kernel Ridge Regression

Ridge regression in feature space:

$$\min_{\boldsymbol{\beta}} \|\boldsymbol{\Phi}\boldsymbol{\beta} - \mathbf{y}\|^2 + \lambda \|\boldsymbol{\beta}\|^2$$

where $\boldsymbol{\Phi} \in \mathbb{R}^{n \times p}$ has rows $\phi(\mathbf{x}_i)^\top$. The solution can be written:

$$\boldsymbol{\beta} = \boldsymbol{\Phi}^\top \boldsymbol{\alpha}$$

for some $\boldsymbol{\alpha} \in \mathbb{R}^n$. Substituting:

$$\boldsymbol{\alpha} = (\mathbf{K} + \lambda \mathbf{I})^{-1} \mathbf{y}$$

where $\mathbf{K} = \boldsymbol{\Phi}\boldsymbol{\Phi}^\top$ is the kernel matrix with $K_{ij} = k(\mathbf{x}_i, \mathbf{x}_j)$.

Prediction at new point $\mathbf{x}$:

$$\hat{y} = \sum_{i=1}^{n} \alpha_i k(\mathbf{x}_i, \mathbf{x}) = \mathbf{k}(\mathbf{x})^\top (\mathbf{K} + \lambda \mathbf{I})^{-1} \mathbf{y}$$

where $\mathbf{k}(\mathbf{x}) = (k(\mathbf{x}_1, \mathbf{x}), \ldots, k(\mathbf{x}_n, \mathbf{x}))^\top$.

Computational Considerations

Kernel methods scale as $O(n^3)$ (matrix inversion) and $O(n^2)$ storage, limiting applicability to $n \lesssim 10^4$. Approximations include:

- Random Fourier features (approximate RBF kernel)
- Nyström approximation (low-rank $\mathbf{K}$)

Gaussian Process Regression

Kernel ridge regression has a Bayesian interpretation as Gaussian Process (GP) regression.

Definition 29 (Gaussian Process)**.** *A **Gaussian Process** is a collection of random variables, any finite subset of which is jointly Gaussian. A GP is specified by:*

- *Mean function:* $m(\mathbf{x}) = \mathbb{E}[f(\mathbf{x})]$
- *Covariance function:* $k(\mathbf{x}, \mathbf{x}') = Cov(f(\mathbf{x}), f(\mathbf{x}'))$

We write $f \sim \mathcal{GP}(m, k)$.

GP Regression

Assume $y_i = f(\mathbf{x}_i) + \varepsilon_i$ with $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$ and prior $f \sim \mathcal{GP}(0, k)$.

The posterior predictive distribution at $\mathbf{x}_*$ is:

$$f_* | \mathbf{X}, \mathbf{y}, \mathbf{x}_* \sim \mathcal{N}(\bar{f}_*, \mathrm{var}(f_*)) \tag{76}$$

with:

$$\bar{f}_* = \mathbf{k}_*^\top (\mathbf{K} + \sigma^2 \mathbf{I})^{-1} \mathbf{y} \tag{77}$$

$$\mathrm{var}(f_*) = k(\mathbf{x}_*, \mathbf{x}_*) - \mathbf{k}_*^\top (\mathbf{K} + \sigma^2 \mathbf{I})^{-1} \mathbf{k}_* \tag{78}$$

The mean prediction matches kernel ridge regression with $\lambda = \sigma^2$. GPs additionally provide uncertainty quantification.

Multiple Linear Regression: Practical Considerations

Multicollinearity

When predictors are highly correlated, $\mathbf{X}^\top \mathbf{X}$ is nearly singular:

- OLS coefficients have high variance
- Coefficients are unstable (sensitive to small data changes)
- Individual t-tests may be non-significant even when joint F-test is significant

Detection: Variance Inflation Factor (VIF):

$$\mathrm{VIF}_j = \frac{1}{1 - R_j^2}$$

where R_j^2 is the R^2 from regressing x_j on all other predictors. VIF > 10 suggests problematic collinearity.

Solutions:

- Remove redundant features
- Principal Component Regression (PCR)
- Ridge regression (stabilizes via regularization)

Principal Component Regression (PCR)

1. Compute PCA of $\mathbf{X}$: $\mathbf{X} = \mathbf{U}\mathbf{D}\mathbf{V}^\top$
2. Keep top k principal components: $\mathbf{Z} = \mathbf{X}\mathbf{V}_k = \mathbf{U}_k\mathbf{D}_k$
3. Regress $\mathbf{y}$ on $\mathbf{Z}$: $\hat{\gamma} = (\mathbf{Z}^\top\mathbf{Z})^{-1}\mathbf{Z}^\top\mathbf{y}$
4. Transform back: $\hat{\boldsymbol{\theta}}_{\text{PCR}} = \mathbf{V}_k\hat{\gamma}$

PCR discards high-variance directions (small singular values), acting as implicit regularization. Unlike ridge, PCR is a hard threshold (keep or discard) rather than soft shrinkage.

Partial Least Squares (PLS)

Like PCR, but constructs components that maximize covariance with $\mathbf{y}$, not just variance in $\mathbf{X}$. Useful when the high-variance directions in $\mathbf{X}$ are not predictive of $\mathbf{y}$.

Heteroskedasticity and Robust Standard Errors

When $\text{Var}(\varepsilon_i) = \sigma_i^2$ varies (heteroskedasticity), OLS remains unbiased but:

- OLS is no longer BLUE (GLS is)
- Standard errors are incorrect $\Rightarrow$ invalid inference

Detection

- **Breusch-Pagan test**: Regress $\hat{\varepsilon}_i^2$ on $\mathbf{x}_i$; significant $\Rightarrow$ heteroskedasticity
- **White test**: Regress $\hat{\varepsilon}_i^2$ on $\mathbf{x}_i$, $\mathbf{x}_i^2$, and cross-products

Heteroskedasticity-Robust Standard Errors

The **sandwich estimator** (White, Huber-White):

$$\widehat{\text{Var}}(\hat{\boldsymbol{\theta}}) = (\mathbf{X}^\top\mathbf{X})^{-1}\left(\sum_{i=1}^{n}\hat{\varepsilon}_i^2\mathbf{x}_i\mathbf{x}_i^\top\right)(\mathbf{X}^\top\mathbf{X})^{-1}$$

This is consistent for $\text{Var}(\hat{\boldsymbol{\theta}})$ under heteroskedasticity.

Generalized Least Squares (GLS)

If $\text{Var}(\boldsymbol{\varepsilon}) = \sigma^2\boldsymbol{\Omega}$ with known $\boldsymbol{\Omega}$:

$$\hat{\boldsymbol{\theta}}_{\text{GLS}} = (\mathbf{X}^\top\boldsymbol{\Omega}^{-1}\mathbf{X})^{-1}\mathbf{X}^\top\boldsymbol{\Omega}^{-1}\mathbf{y}$$

GLS is BLUE under heteroskedasticity/autocorrelation. Feasible GLS (FGLS) estimates $\boldsymbol{\Omega}$ from the data.

Quantile Regression

OLS estimates the conditional mean $\mathbb{E}[y|\mathbf{x}]$. **Quantile regression** estimates conditional quantiles.

Definition 30 (Quantile Regression)**.** *The τ-th conditional quantile ($\tau \in (0,1)$) is estimated by:*

$$\hat{\boldsymbol{\theta}}_\tau = \arg\min_{\boldsymbol{\theta}} \sum_{i=1}^{n} \rho_\tau(y_i - \mathbf{x}_i^\top \boldsymbol{\theta})$$

where $\rho_\tau(u) = u(\tau - \mathbf{1}_{u<0})$ is the check function.

For $\tau = 0.5$, this is median regression (minimizes absolute deviations). Quantile regression is robust to outliers and provides a richer picture of the conditional distribution.

Summary of Methods

Method	**Penalty**	**Sparsity**	**Closed Form**	**When to Use**
OLS	None	No	Yes	Baseline, $n \gg d$
Ridge	$\lambda\|\boldsymbol{\theta}\|_2^2$	No	Yes	Multicollinearity, d large
Lasso	$\lambda\|\boldsymbol{\theta}\|_1$	Yes	No	Feature selection
Elastic Net	$\lambda_1\|\cdot\|_1 + \lambda_2\|\cdot\|_2^2$	Yes	No	Groups of correlated features
PCR	Implicit	No	Yes	High-dim, variance reduction
Kernel Ridge	$\lambda\|\boldsymbol{\beta}\|^2$ in $\mathcal{H}$	No	Yes	Nonlinear relationships